Practice workbook · Logical and arithmetic binary shifts

Logical and arithmetic shifts left or right by n places, and the multiply/divide effect of a shift.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-07

1. Question 2 · part c.ii · 1 mark

2023 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

P 7 2 9 3 7 A 0 5 2 4

1. Question 2 · part c.ii · 1 mark

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Worked-example retry · Attempt part c.ii only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

1. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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2. Question 1 · part e · 1 mark

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Further practice · Attempt part e only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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2. Question 1 · part e · 1 mark

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Further practice · Attempt part e only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

(f) Give the number of bits per character used by standard ASCII.

(1)

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

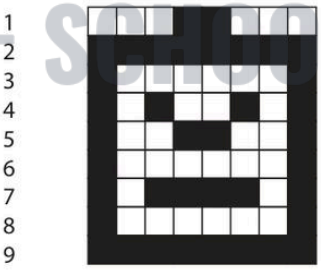


Figure 1

(i) State what is meant by the term **pixel**.

(1)

(ii) Construct an expression to show the number of pixels in the image.

(1)

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2. Question 1 · part e · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.i · 1 mark

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Further practice · Attempt part c.i only.

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(1)

.....

.....

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

.....

.....

P 7 2 9 3 7 A 0 5 2 4

5

Turn over ▶

3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

3 Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

(1)

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

(3)

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

(1)

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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4. Question 3 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

(2)

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

(1)

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

(2)

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4. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

5. Question 3 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

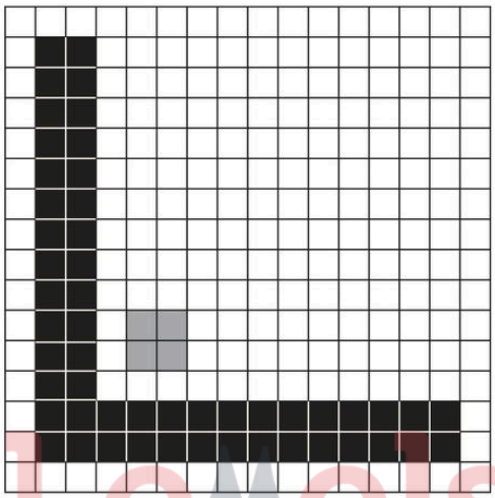


Figure 1

(i) State the smallest component of a bitmap image.

(1)

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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5. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

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